Chapter 8 Practice Worksheet

Exponents, Radicals & Exponential Functions

Name: _____ Date: _____ Period: _____

Key Formulas: $a^m \cdot a^n = a^{m+n}$ $(a^m)^n = a^{mn}$ $\dfrac{a^m}{a^n} = a^{m-n}$ $a^0 = 1$ $a^{-n} = \dfrac{1}{a^n}$ $a^{m/n} = \sqrt[n]{a^m}$

Part A Laws of Exponents (10 problems)

Simplify each expression completely. Leave all answers with positive exponents.

A1–A5 *Product Rule, Power Rule, Zero & Negative Exponent*

A1. $3^2 \cdot 3^5$

A2. $(5^2)^3$

A3. 7^0

A4. 4^{-3}

A5. $(2^4)^2 \cdot 2^{-3}$

A6–A10 *Quotient Rule, Multi-step Simplification*

A6. $\dfrac{x^8}{x^3}$

A7. $(3x^2)^4$

A8. $\dfrac{a^{-2} \cdot a^5}{a^3}$

A9. $(2^3 \cdot 3^2)^0$

A10. $\dfrac{(2x^3)^2}{x^4}$

Part B Radicals & Rational Exponents (10 problems)

Convert between radical and rational-exponent form, or evaluate the expression.

B1–B5 *Evaluate using Radicals & Rational Exponents*

B1. Evaluate: $\sqrt[10]{1024}$

B2. Write as a radical, then evaluate: $8^{1/3}$

B3. Evaluate: $27^{2/3}$ *Hint: find the cube root first, then square.*

B4. Evaluate: $16^{3/4}$ *Hint:* $16 = 2^4$.

B5. Evaluate: $\sqrt[5]{32}$

B6–B10 *Convert Between Forms*

B6. Write as a rational exponent: $\sqrt[4]{x^3}$

B7. Write in radical notation: $x^{5/6}$

B8. Evaluate: $(-8)^{1/3}$

B9. Simplify: $\dfrac{1}{9^{1/2}}$

B10. Simplify: $(4x^2)^{3/2}$ *Hint: apply the rational exponent to each factor.*

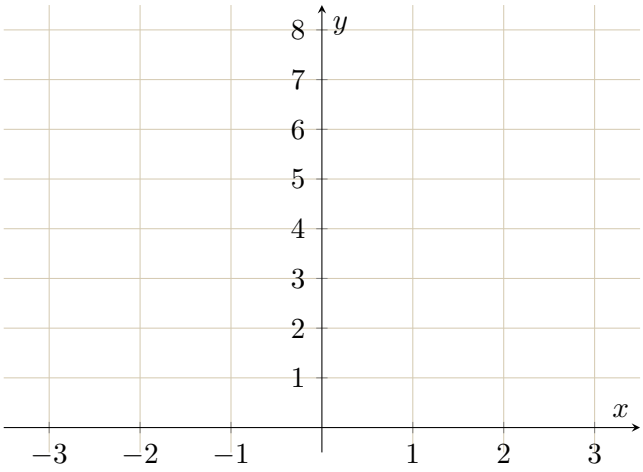
Part C The Exponential Function $y = 2^x$ (10 problems)

Answer each question about the exponential function $y = 2^x$.

C1. Table of values. Complete the table for $y = 2^x$:

x	-3	-2	-1	0	1	2	3
$y = 2^x$							

C2. Sketch the graph. Using your table above, graph $y = 2^x$ below. Label at least **four** points clearly.



C3. Domain & Range. State the domain and range of $y = 2^x$.

Domain: _____ Range: _____

C4. y-intercept. What is the y-intercept of $y = 2^x$? Explain why.

C5. x-intercept. Does $y = 2^x$ have an x-intercept? Explain your answer.

C6. End behavior. Complete the sentences below.

As $x \to -\infty$, $y \to$ _____ because _____

As $x \to +\infty$, $y \to$ _____ because _____

C7. Horizontal asymptote. True or False: $y = 2^x$ has a horizontal asymptote at $y = 0$.
Circle one: **TRUE** / **FALSE**

Explain: _____

C8. Solve for x. If $2^x = 64$, what is x?

Hint: rewrite 64 as a power of 2.

C9. Linear vs. Exponential. The table below shows values for $y = 2x$ (linear) and $y = 2^x$ (exponential). Fill in the missing values, then circle which function grows faster for large x.

x	$y = 2x$	$y = 2^x$
1		
5		
10		
20		

Faster growth for large x: $y = 2x$ / $y = 2^x$ (circle one)

C10. Connect to Radicals. The point (10, 1024) lies on $y = 2^x$.

(a) Express this using *radical notation*: $\sqrt{\quad} =$ _____

(b) Express this using a *rational exponent*: ( ) = _____

(c) Verify your answer to (b) using the Laws of Exponents. Show your work:

Answer Key

Chapter 8 Exponents, Radicals & Exponential Functions

Part A Laws of Exponents

A1 $3^2 \cdot 3^5 = 3^7 = \mathbf{2187}$

A2 $(5^2)^3 = 5^6 = \mathbf{15{,}625}$

A3 $7^0 = \mathbf{1}$

A4 $4^{-3} = \dfrac{1}{4^3} = \mathbf{\dfrac{1}{64}}$

A5 $(2^4)^2 \cdot 2^{-3} = 2^8 \cdot 2^{-3} = 2^5 = \mathbf{32}$

A6 $\dfrac{x^8}{x^3} = \mathbf{x^5}$

A7 $(3x^2)^4 = 3^4\, x^8 = \mathbf{81x^8}$

A8 $\dfrac{a^{-2} \cdot a^5}{a^3} = a^{-2+5-3} = a^0 = \mathbf{1}$

A9 $(2^3 \cdot 3^2)^0 = \mathbf{1}$
Any nonzero base to the 0th power = 1.

A10 $\dfrac{(2x^3)^2}{x^4} = \mathbf{4x^2}$

Part B Radicals & Rational Exponents

B1 $\sqrt[10]{1024} = \mathbf{2}$
since $2^{10} = 1024$

B2 $8^{1/3} = \sqrt[3]{8} = \mathbf{2}$

B3 $27^{2/3} = \left(\sqrt[3]{27}\right)^2 = 3^2 = \mathbf{9}$

B4 $16^{3/4} = \left(\sqrt[4]{16}\right)^3 = 2^3 = \mathbf{8}$

B5 $\sqrt[5]{32} = \mathbf{2}$
since $2^5 = 32$

B6 $\sqrt[4]{x^3} = \mathbf{x^{3/4}}$

B7 $x^{5/6} = \sqrt[6]{x^5}$

B8 $(-8)^{1/3} = \mathbf{-2}$
since $(-2)^3 = -8$

B9 $\dfrac{1}{9^{1/2}} = \dfrac{1}{\sqrt{9}} = \mathbf{\dfrac{1}{3}}$

B10 $(4x^2)^{3/2} = 4^{3/2} \cdot x^3 = 8x^3 \;\Rightarrow\; \mathbf{8x^3}$

Part C The Exponential Function $y = 2^x$

C1. Table of values:

x	-3	-2	-1	0	1	2	3
$y = 2^x$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8

C2. Graph should pass through all seven points above, rising steeply to the right and flattening toward $y = 0$ to the left.

C3. Domain: $(-\infty, +\infty)$; Range: $(0, +\infty)$

C4. y-intercept: $(0, 1)$, because $2^0 = 1$.

C5. No x-intercept. $2^x > 0$ for all real x, so the curve never crosses the x-axis.

C6. As $x \to -\infty$: $y \to 0$ (approaches asymptote $y = 0$). As $x \to +\infty$: $y \to +\infty$ (exponential growth).

C7. TRUE. The line $y = 0$ (the x-axis) is a horizontal asymptote.

C8. $2^x = 64 = 2^6 \Rightarrow x = \mathbf{6}$

C9. Completed table and faster-growth answer:

x	$y = 2x$	$y = 2^x$
1	2	2
5	10	32
10	20	1,024
20	40	1,048,576

Faster growth: $\mathbf{y = 2^x}$ (exponential).

C10. (a) $\sqrt[10]{1024} = 2$ (b) $(1024)^{1/10} = 2$ (c) $(2^{10})^{1/10} = 2^{10 \cdot \frac{1}{10}} = 2^1 = 2$ ✓